



Cambridge IGCSE™

BIOLOGY

0610/63

Paper 6 Alternative to Practical

May/June 2021

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **7** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- | | |
|---|--|
| 1 | Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly. |
| 2 | The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored. |
| 3 | Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection). |
| 4 | The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted. |
| 5 | <p><u>'List rule' guidance</u></p> <p>For questions that require <i>n</i> responses (e.g. State two reasons ...):</p> <ul style="list-style-type: none"> • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science. |

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks	Guidance																							
1(a)(i)	0.75 ;	1																								
1(a)(ii)	table with minimum two columns and a header row ; suitable headings ; additional columns for two trials and average with appropriate headings ; eight values recorded ; four correct averages calculated ;	5	e.g. <table> <tr> <th rowspan="2">percentage concentration of vitamin C</th><th colspan="3">number of drops of DCPIP</th></tr> <tr> <th>trial 1</th><th>trial 2</th><th>average</th></tr> <tr> <td>0.25</td><td>6</td><td>4</td><td>5</td></tr> <tr> <td>0.50</td><td>13</td><td>11</td><td>12</td></tr> <tr> <td>0.75</td><td>18</td><td>18</td><td>18</td></tr> <tr> <td>1.00</td><td>23</td><td>21</td><td>22</td></tr> </table>	percentage concentration of vitamin C	number of drops of DCPIP			trial 1	trial 2	average	0.25	6	4	5	0.50	13	11	12	0.75	18	18	18	1.00	23	21	22
percentage concentration of vitamin C	number of drops of DCPIP																									
	trial 1	trial 2	average																							
0.25	6	4	5																							
0.50	13	11	12																							
0.75	18	18	18																							
1.00	23	21	22																							
1(a)(iii)	higher concentrations of vitamin C require greater number of drops of DCPIP ; ora	1																								
1(a)(iv)	to identify anomalous results / AW ;	1																								
1(a)(v)	a value in range of 0.26 to 0.50 (%) ;	1																								
1(a)(vi)	<i>any two from:</i> repeat the measurement for H ; use intermediate concentrations / AW ; measure the volume of DCPIP instead of drops ; use a larger volume of vitamin C / lower DCPIP concentration ; shake after adding each drop ; keep temperature constant ; AVP ; e.g. plot, results / values, on a graph and read off value for 10 drops from a line of best fit	2																								
1(a)(vii)	<i>any two from:</i> concentration of DCPIP ; volume of vitamin C solution ; type of indicator ;	2																								

Question	Answer	Marks	Guidance
1(b)	<i>any one from:</i> drop size is variable / unknown drop size ; unable to determine exact end-point as it may be between two drops / AW ;	1	
1(c)	<i>protein</i> biuret (solution / reagent) ; purple colour ; <i>glucose</i> Benedict's (solution / reagent) ; heat ; (brick-)red / orange / yellow / green, colour ;	5	

Question	Answer	Marks	Guidance
2(a)(i)	Size – more than 50% of answer gap used ; Quality – continuous outline, no shading ; Details ;; alternating pattern of veins / 4 to 10 veins drawn on upper half of leaf	4	
2(a)(ii)	measurement of line AB as 40 ± 1 mm ; line AB drawn on candidate's drawing in 2(a)(i) ; magnification calculated from measured values ;	3	
2(b)(i)	axes labelled with units ; linear scale for plotted points to occupy half or more of the grid in both dimensions ; <u>all</u> plotted points accurate to $\pm$ half small square ; line with no extrapolation and points joined with a ruler ;	4	(number of) bubbles in 5 minutes + carbon dioxide concentration, mg per dm ³ A line of best fit
2(b)(ii)	as carbon dioxide increases, number of bubbles increases (and levels off) / ora ;	1	

Question	Answer	Marks	Guidance
2(b)(iii)	3 (bubbles per minute) ;;	2	MP1 correct rate calculation i.e. $14 \div 5 = 2.8$ (bubbles per minute) MP2 correct rounding to one significant figure
2(b)(iv)	carbon dioxide concentration ;	1	
2(c)	<i>any six from:</i> <i>independent variable:</i> 1 at least two light intensities stated ; <i>given method:</i> 2 set up lamp with plant in water ; <i>new method:</i> 3 use a ruler to measure distance of lamp ; 4 change distance of lamp / bulb wattage / AW ; 5 use of a heat shield ; 6 ref. to equilibration time ; 7 ref. suitable apparatus for collecting gas e.g. gas syringe / (inverted) measuring cylinder / photosynthometer ; 8 measure volume in set time / record time to produce set volume ; 9 & 10 <i>controlled variables</i> ;; <i>max two from:</i> temperature type / species, of plant number of leaves / leaf surface area / size / mass pH carbon dioxide concentration volume of water 11 relevant safety precaution ; e.g. ref. electric lamp and water 12 repeat the whole investigation at least two more times ;	6	